EXPERIMENT – 2 : Colour Models and Visualisation of Colour Components

Aim

To separate and visualise the Red, Green and Blue components of a colour image and represent the image in CMY, HSV and YCbCr colour models using MATLAB.

Software Required

- Personal computer
- MATLAB R2020a or later
- Image Processing Toolbox

Theory

RGB Model: RGB is an additive colour model that combines red, green and blue light to produce different colours. It is commonly used in cameras and displays.

CMY Model: CMY is a subtractive colour model used in printing. It is obtained by subtracting the normalized RGB values from 1.

\[C=1-R,\quad M=1-G,\quad Y=1-B \]

HSV Model: HSV represents colour using Hue, Saturation and Value. It is useful for colour enhancement and image segmentation.

YCbCr Model: YCbCr separates luminance (Y) from chrominance components (Cb and Cr). It is widely used in image and video compression.

Program

```
clc;

clear;

close all;

% Read colour image

RGB = imread('peppers.png');

% Extract RGB components

R = RGB(:,:,1);

G = RGB(:,:,2);

B = RGB(:,:,3);
```

```
Z = zeros(size(R), 'uint8');
% RGB component visualisation
figure('Name','RGB Components');
subplot(2,4,1);
imshow(RGB);
title('Original Image');
subplot(2,4,2);
imshow(cat(3,R,Z,Z));
title('Red Component');
subplot(2,4,3);
imshow(cat(3,Z,G,Z));
title('Green Component');
subplot(2,4,4);
imshow(cat(3,Z,Z,B));
title('Blue Component');
subplot(2,4,6);
imshow(R);
title('R Intensity');
subplot(2,4,7);
imshow(G);
title('G Intensity');
subplot(2,4,8);
imshow(B);
title('B Intensity');
% CMY colour model
CMY = imcomplement(RGB);
figure('Name','CMY Model');
subplot(1,4,1);
```

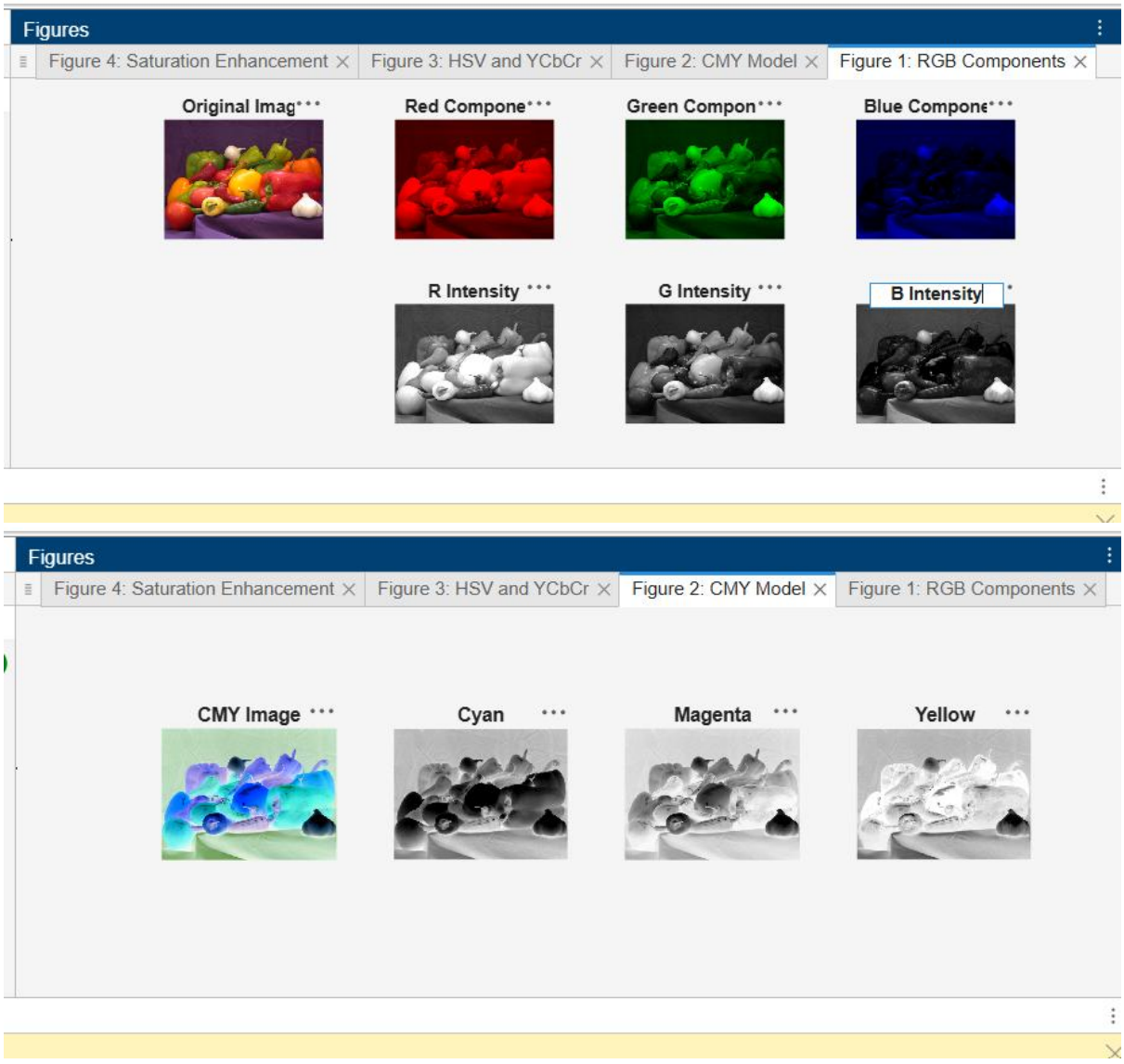
```
imshow(CMY);
title('CMY Image');
subplot(1,4,2);
imshow(CMY(:,:,1));
title('Cyan');
subplot(1,4,3);
imshow(CMY(:,:,2));
title('Magenta');
subplot(1,4,4);
imshow(CMY(:,:,3));
title('Yellow');
% HSV and YCbCr models
HSV = rgb2hsv(RGB);
YCC = rgb2ycbcr(RGB);
figure('Name','HSV and YCbCr');
subplot(2,3,1);
imshow(HSV(:,:,1));
title('Hue');
subplot(2,3,2);
imshow(HSV(:,:,2));
title('Saturation');
subplot(2,3,3);
imshow(HSV(:,:,3));
title('Value');
subplot(2,3,4);
imshow(YCC(:,:,1));
title('Y Luminance');
subplot(2,3,5);
```

```
imshow(YCC(:,:,2));
title('Cb');
subplot(2,3,6);
imshow(YCC(:,:,3));
title('Cr');
% Display one pixel in different models
r = 200;
c = 300;
fprintf('Pixel (%d,%d)\n', r, c);
fprintf('RGB = [%d %d %d]\n', RGB(r,c,1), ...
   RGB(r,c,2), RGB(r,c,3));
fprintf('CMY = [%d %d %d]\n', CMY(r,c,1), ...
   CMY(r,c,2), CMY(r,c,3));
fprintf('HSV = [%.3f %.3f %.3f]\n', HSV(r,c,1), ...
   HSV(r,c,2), HSV(r,c,3));
fprintf('YCbCr = [%d %d %d]\n', YCC(r,c,1), ...
   YCC(r,c,2), YCC(r,c,3));
% Saturation enhancement
HSV2 = HSV;
HSV2(:,:,2) = min(1.5*HSV(:,:,2), 1);
figure('Name','Saturation Enhancement');
subplot(1,2,1);
imshow(RGB);
title('Original Image');
subplot(1,2,2);
imshow(hsv2rgb(HSV2));
title('Saturation Increased');
```

Procedure

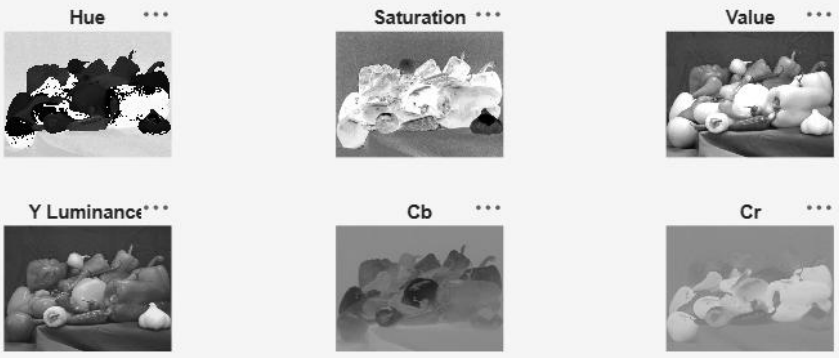
1. Open MATLAB and set the Current Folder to your working directory.
2. Create a new script and save it as exp02.m.
3. Enter the program and execute it by pressing F5.
4. Observe the RGB components, CMY planes, HSV components and YCbCr components.
5. Record the RGB, CMY, HSV and YCbCr values for the selected pixel.
6. Change the pixel coordinates to examine different coloured peppers.

Output



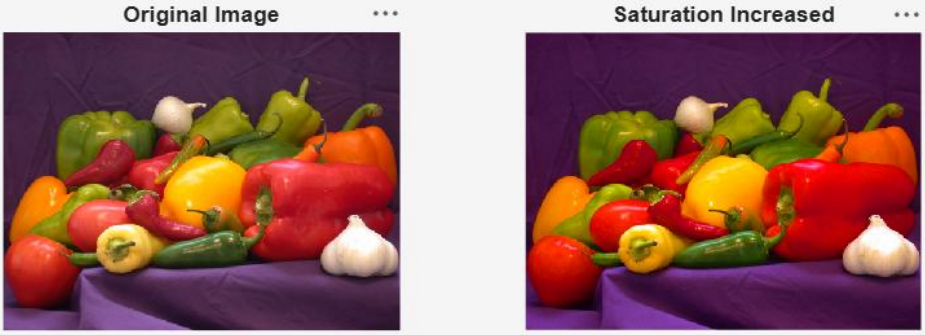
Figures

Figure 4: Saturation Enhancement | Figure 3: HSV and YCbCr | Figure 2: CMY Model | Figure 1: RGB Components



Figures

Figure 4: Saturation Enhancement | Figure 3: HSV and YCbCr | Figure 2: CMY Model | Figure 1: RGB Components



Command Window

New to MATLAB? See resour

```
Pixel (200,300)
RGB = [255 150 0]
CMY = [0 105 255]
HSV = [0.098 1.000 1.000
YCbCr = [157 47 185]
```

Observations

Pixel (r, c)	RGB	CMY	HSV	YCbCr
(200, 300)	Record output	Record output	Record output	Record output
Red pepper pixel	Record output	Record output	Record output	Record output
Green pepper pixel	Record output	Record output	Record output	Record output
Yellow pepper pixel	Record output	Record output	Record output	Record output

Note: The pixel values depend on the selected image and coordinates.

Result

The RGB colour components were successfully separated and visualised. The image was converted into CMY, HSV and YCbCr colour models, and saturation enhancement was performed using MATLAB.